

Sarthak Avhad

+91-79721 43404 | avhadsarthak908@gmail.com | [LinkedIn](#) | [GitHub](#)

PROFESSIONAL SUMMARY

M.Sc. Computer Science student specializing in Blockchain Technology at MIT World Peace University, with hands-on project experience spanning blockchain infrastructure (Solidity, Ethereum, zk-SNARKs, Solana, DeFi), full-stack web development (MERN stack), and DevOps/CI-CD workflows. Comfortable building AI-integrated applications end-to-end — from smart contract and backend logic to automated testing and deployment pipelines. Deeply interested in decentralized systems and privacy-preserving protocols.

PROJECTS

- TestGen-AI** — *FastAPI, Python, OpenAI, Gemini, Supabase* Jun. 2026
- **AI Test Generation** Built a full-stack platform that generates unit tests from Python source code via AST analysis and natural language user stories, powered by GPT-4.1-mini with Gemini fallback.
 - **Execution Engine** Integrated `pytest` and `pytest-cov` via FastAPI to execute tests in isolated environments, returning pass/fail status and line-level coverage reports.
 - **Analytics Pipeline** Persisted generated tests, execution logs, and coverage reports in Supabase (PostgreSQL), powering a historical analytics dashboard.
- StealthPay** — *MPC, Web3, Node.js, EIP-5564, secp256k1* Mar. 2026
- **Privacy-Preserving Payments** Built an Ethereum dApp implementing EIP-5564 stealth addresses with a Next.js frontend and Node.js blockchain scanner.
 - **Protocol Engineering** Implemented ECDH-based key derivation, stealth meta-address registry support, and ephemeral key handling compliant with EIP-5564/EIP-6538.
 - **Recognition** Received recognition from BitGo engineers at EthMumbai 2026 for the stealth address protocol implementation.
- LinkStore** — *Next.js, TypeScript, Supabase, Cheerio* Jan. 2026
- **Link Management** Developed a full-stack bookmark manager with authenticated CRUD operations, category organization, and Row-Level Security using Supabase.
 - **Metadata Engine** Built a resilient Open Graph metadata scraper using Cheerio, with Twitter Card and HTML fallbacks for rich link previews.
 - **Performance & Security** Implemented multi-level sorting, pinning, XSS sanitization with DOMPurify, and strict Content Security Policy headers for secure rendering.

EDUCATION

- MIT World Peace University** Jun. 2025 – Aug. 2027
M.Sc. in Computer Science (Blockchain Technology) Pune, Maharashtra
- K.M.C. College of Arts, Science & Commerce** Jun. 2022 – Apr. 2025
Bachelor of Computer Science (BSC-CS) – CGPA: 8.5 Khopoli, Maharashtra

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, C++
Frontend: React, Next.js, Tailwind CSS, shadcn/ui
Backend: Node.js, FastAPI, Express, REST APIs, Microservices, JWT Authentication
Databases: PostgreSQL, Supabase, MongoDB, Prisma ORM, pgvector
AI & Agents: RAG, LLM Integration, n8n, MCP
Blockchain & Cryptography: Ethereum, Solidity, ethers.js, zkSync Era, Bitcoin, Stealth Addresses, Wallet Integration, zkSNARKs, ECDH, Dilithium, Post-Quantum Cryptography
Tools & Platforms: Linux, Git, Docker, GitHub Actions, YAML, Vercel, Nosana CLI, Hugging Face, Postman